Static Voltage Stability Analysis of Renewable Energy Multi-Infeed Systems Based on Margin Index

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***Abstract*—Substituting a high proportion of renewable energy sources for traditional synchronous generators can lead to a deterioration in the system's static voltage stability. This paper proposes extending margin-based voltage stability indices based on the load side Thevenin equivalent, to the renewable energy source side for assessing the static voltage stability of renewable energy multi-infeed systems. Firstly, a comprehensive analysis of both the load and renewable energy sources is conducted, and normalized expressions for three margin indices are derived based on a single-machine black-box system. Secondly, the characteristics of the margin indices are analyzed, validating their applicability in a single-infeed system. Thirdly, a method is introduced for conducting static stability analysis on the renewable energy source side of multi-infeed systems using Thevenin equivalent. Finally, the effectiveness of the indices under different growth scenarios is verified in a modified IEEE 39-node system, expanding the application scenarios of these indices.**

***Keywords*—static voltage stability; margin index; renewable energy multi-infeed system; Thevenin equivalent; sensitivity**

I. INTRODUCTION

In recent years, China has accelerated the construction of a new type of power system. The total installed capacity and annual power generation of renewable energy units have experienced rapid growth [1-2]. However, accompanying this growth is a gradual weakening of the voltage stability of the power system. Taking the Jixi renewable energy base as an example, with the continued integration of renewable energy sources in the western region of Jilin in 2022, the short-circuit ratio of multiple renewable energy stations in some areas will fall below 1.5, failing to meet the requirements for the voltage strength of multiple renewable energy stations [3]. This situation could easily lead to widespread disconnection of renewable energy sources. Therefore, accurately assessing the static voltage stability of the renewable energy transmission system holds significant importance for the safe and stable operation of the system.

The evaluation indices for static voltage stability primarily fall into two categories: margin indices and state indices. Common state indices include sensitivity indices [4], energy function indices [5], eigenvalue indices [6], and others. Taking sensitivity indices as an example, as the system approaches instability, the sensitivity of voltage to power tends to approach infinity. This characteristic is used as a basis for determining whether the system is unstable or approaching instability. However, due to the poor linearity of state indices, especially in the distance from the turning point to the instability point, the values of state indices can change rapidly. This makes it challenging to timely and accurately predict the voltage stability margin.

Margin indices represent a widely used type of assessment metric employed in analyzing static voltage stability. These indices involve setting a specific growth pattern to compare the distance between voltage, power, and other metrics in the current system state and the critical stability state. This comparison is pivotal in evaluating the voltage stability of a system. These indices offer straightforward calculations and clear physical interpretations, crucial in static voltage stability analysis. Among them, the impedance margin does not rely on the calculation of the Thevenin equivalent potential. Its application has been extensive, with abundant research findings. Literature [7-8] independently validated the efficacy of impedance margin indices in photovoltaic integration systems and radiative distribution networks. However, most existing research has predominantly focused on the load side. In literature [9], there's a proposal to extend impedance margin indices to the realm of renewable energy sources. The study conducted stability analyses of wind power stations based on these indices, yielding certain results. However, this research didn't consider the strong nonlinearity of impedance margin indices near instability points. "Moreover, the accuracy of its Thevenin equivalent method based on node admittance matrix remains questionable under strong nonlinear conditions."

The Short Circuit Ratio (SCR) is another type of margin index. Traditionally defined as the ratio between the system's short-circuit capacity and the equipment's rated capacity [10]. With renewable energy stations employing multi-infeed methods, their interdependencies impact each other. Calculating based on a single-infeed method can yield overly optimistic results. Hence, numerous scholars have researched to improve the SCR index [11-13]. Based on operational experiences with renewable energy generation worldwide, the critical SCR is generally around 1.5 [3]. However, during actual operation, factors like system parameters and operating conditions significantly influence the critical SCR, which cannot be overlooked. Literature [12] presents various methods to calculate critical values for multiple SCR indices. However, these indices are applicable in different scenarios, varying in precision, which poses challenges for practical application.

To address the difficulty in calculating critical values for common short-circuit ratio indices, this paper conducts a comprehensive study on both the load and renewable energy sources. The proposed method establishes a static stability analysis model for a single-machine black-box system, extends the margin indices on the load side, and proposes a new

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method for static stability analysis on the renewable energy source side based on margin indices. This method is employed to evaluate the static voltage stability of renewable energy multi-infeed systems. The main contributions are as follows:

i) By establishing a static stability analysis model for a single-machine black-box system, specific expressions for three margin indices on the load side and renewable energy source side are derived. The proposed model illustrates that the static stability analysis process based on margin indices for both the source and load sides is essentially consistent.

ii) On the basis of stability analysis in a single-infeed system, this paper proposes the simplification of a multi-infeed system using Thevenin equivalent. Subsequently, margin indices are applied to assess the static stability of the renewable energy source side in the multi-infeed system

II. The Power Flow Equations of a Single-Machine Black-Box System

The control strategy of grid-connected inverters for renewable energy sources has a significant impact on the static voltage stability at the grid connection point, mainly employing two methods: constant power factor control and constant voltage control. In static stability analysis, the renewable energy station under constant voltage control can be regarded as a PV node. There are no instability issues when the reactive power remains within limits. Conversely, renewable energy stations employing constant power factor control cannot establish sufficient voltage support and are thus regarded as PQ nodes. This study focuses on renewable energy stations utilizing constant power factor control.

In static stability analysis, neglecting load static characteristics, the only difference between renewable energy stations and loads lies in the power direction. Thus, considering a unified description for both, they can be represented by a single black-box element. Specifically, performing a Thevenin equivalent on the system side of the grid at the black-box side yields the single-machine black-box system shown in Fig 1.

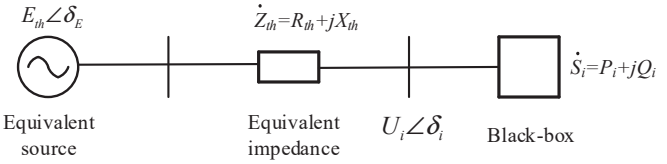


Fig. 1. Single-machine black-box system

The given reference direction for power is from the system side towards the black box. Based on the fundamental equations of the circuit, the power flow equations are formulated as follows [14]:

$$\begin{cases} S_i\cos\varphi=\dfrac{U_i}{Z_{th}}\left[E_{th}\cos(\theta-(\delta_E-\delta_i))-U_i\cos\theta\right] \\ S_i\sin\varphi=\dfrac{U_i}{Z_{th}}\left[E_{th}\sin(\theta-(\delta_E-\delta_i))-U_i\sin\theta\right] \end{cases} \quad (1)$$

In (1): S_i represents the apparent power magnitude at node i; U_i represents the voltage magnitude at node i; δ_i represents the voltage phase angle at node i; Eth represents the magnitude of Thevenin equivalent source; δ_E represents the phase angle of Thevenin equivalent source; Z_{th} represents the magnitude of Thevenin equivalent impedance; θ represents the angle of Thevenin equivalent impedance; φ represents the power factor angle of the black-box element. By simultaneously eliminating (δ_E-δ_i), an analytical expression describing the relationship between the voltage and power at nodes in the depicted single-machine black-box system can be obtained.

$$\left(U_i^2\right)^2+[2Z_{th}S_i\cos(\theta-\varphi)-E_{th}^2]U_i^2+Z_{th}^2S_i^2=0 \quad (2)$$

In (2), the Thevenin equivalent parameters E_{th}, Z_{th}, θ, and the power factor angle φ of the black-box element are all known quantities. Solving (2) allows us to determine the power flow state of the system. If power flows from the system side towards the black box, then the black box represents a load. Conversely, if power flows from the black box towards the system side, then the black box represents a renewable energy station.

III. The Expression for Voltage Stability Margin Index on the Source and Load Sides

A. The expression for the voltage margin index K_U on the source and load sides

The voltage margin index K_U is the simplest index to determine whether the voltage at a certain node is on the verge of collapse. This index directly reflects the voltage at the node, providing a clear physical meaning. Taking U_i^2 as the unknown in solving (2) as a system of quadratic equations, the expression for the node voltage of the black box component can be obtained as follows:

$$U_i=\sqrt{\frac{E_{th}^2}{2}-Z_{th}S_i\cos(\theta-\varphi)\pm A} \quad (3)$$

$$A=\sqrt{\frac{E_{th}^4}{4}-(Z_{th}S_i\sin(\theta-\varphi))^2-E_{th}^2Z_{th}S_i\cos(\theta-\varphi)} \quad (4)$$

Due to the multiple solutions of the power flow equation, theoretically, there are two solutions for voltage, corresponding to the stable high-voltage solution and the unstable low-voltage solution. When these two solutions are equal, that is, when (2) has a repeated root, it is considered that the system is at the voltage stability limit, corresponding precisely to the nose point on the P-V curve. At this point, the discriminant of the quadratic equation concerning U_i^2 is zero:

$$E_{th}^4-4\left(Z_{th}S_i\right)^2\sin(\theta-\varphi)-4E_{th}^2Z_{th}S_i\cos(\theta-\varphi)=0 \quad (5)$$

Solving this quadratic equation with Si as the unknown, we can obtain both positive and negative solutions, corresponding to two scenarios inside the black box: one for the load and the other for the renewable energy source, as shown in equation (6)

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$$S_i = \frac{\pm E_{th}^2 - E_{th}^2 \cos(\theta - \varphi)}{2Z_{th} \sin^2(\theta - \varphi)} \tag{6}$$

When $S_i > 0$, indicating that there is a load inside the black box, substituting (6) into (3), we obtain the critical voltage for the load node:

$$U_{cr.Li} = \frac{E_{th}}{\sqrt{2[1 + \cos(\theta - \varphi)]}} \tag{7}$$

The normalized voltage margin index for the load node is denoted as $K_{U.Li}$ can be expressed as:

$$K_{U.Li} = \frac{U_i - U_{cr.Li}}{U_i} = 1 - \frac{E_{th}}{U_i \sqrt{2[1 + \cos(\theta - \varphi)]}} \tag{8}$$

When $S_i < 0$, indicating that there is a renewable energy generation station inside the black box, substituting (6) into (3), we obtain the critical voltage for the renewable energy node:

$$U_{cr.Wi} = \frac{E_{th}}{\sqrt{2[1 - \cos(\theta - \varphi)]}} \tag{9}$$

The normalized voltage margin index for the renewable energy node is denoted as $K_{U.Wi}$ can be expressed as:

$$K_{U.Wi} = \frac{U_i - U_{cr.Wi}}{U_i} = 1 - \frac{E_{th}}{U_i \sqrt{2[1 - \cos(\theta - \varphi)]}} \tag{10}$$

Combining (8) and (10), it can be observed that when the node voltage of the black-box component is greater than the critical stability voltage, the node voltage is stable. In this case, $0 < K_{U.i} < 1$, and a larger $K_{U.i}$ indicates a greater margin of voltage stability. At the limit of voltage stability, $K_{U.i} = 0$. It is important to note that, as indicated by (7) and (9), the critical voltage $U_{cr.Li}$ for the load node is smaller than the critical voltage $U_{cr.Wi}$ for the renewable energy node. Additionally, the larger the angle θ of the Thevenin equivalent impedance, the smaller the power factor angle φ for the load (renewable energy), and the closer the two critical voltages are.

B. The expression for the power margin index K_S in terms of the source and load

When the system operates at the critical point of voltage stability, the power transfer between the system and the black box reaches its maximum. By rearranging (6), we can obtain the maximum power transferred from the system to the load $S_{cr.Li}$ and the maximum power transferred from the renewable energy source to the system $S_{cr.Wi}$:

$$S_{cr.Li} = \frac{E_{th}^2}{2Z_{th}[1 + \cos(\theta - \varphi)]} \tag{11}$$

$$S_{cr.Wi} = \frac{E_{th}^2}{2Z_{th}[1 - \cos(\theta - \varphi)]} \tag{12}$$

The normalized power margin index is given by:

$$K_{S,Li} = \frac{S_{cr.Li} - S_i}{S_{cr.Li}} = 1 - \frac{2S_i Z_{th}[1 + \cos(\theta - \varphi)]}{E_{th}^2} \tag{13}$$

$$K_{S,Wi} = \frac{S_{cr.Wi} - S_i}{S_{cr.Wi}} = 1 - \frac{2S_i Z_{th}[1 - \cos(\theta - \varphi)]}{E_{th}^2} \tag{14}$$

Combining (13) and (14), it can be observed that when the power transferred between the system and the black box is less than the maximum transfer power, the voltage at that node is stable. At this point, $0 < K_{S.i} < 1$, and a larger value of $K_{S.i}$ indicates a greater margin of voltage stability. At the limit of voltage stability, $K_{S.i} = 0$.

In practical operation, researchers more commonly use the margin of active power to represent power margin. To ensure consistency in results, the load growth is typically set as constant power factor growth in static stability analysis.

C. The expression of the impedance margin index Kr between the source and the load

Ignoring the dynamic characteristics of the load and renewable energy sources, the equivalent impedance Z_L (Z_W) of the load (renewable energy) can be calculated based on the voltage and power of the node. The equivalent impedance is given by:

$$Z_{L,i} = \left| \frac{\dot{U}_{L,i}}{\dot{I}_{L,i}} \right| = \frac{U_{L,i}^2}{S_{L,i}} \tag{15}$$

$$Z_{W,i} = \left| -\frac{\dot{U}_{W,i}}{\dot{I}_{W,i}} \right| = \frac{U_{W,i}^2}{S_{W,i}} \tag{16}$$

When the system is in a critical stable state, obtained from (7), (11), (15) and (9), (12), (16) respectively:

$$\left| Z_{cr.Li} \right| = \left| \frac{(E_{th} / \sqrt{2[1 + \cos(\theta - \varphi)]})^2}{E_{th}^2 / 2Z_{th}[1 + \cos(\theta - \varphi)]} \right| = \left| Z_{th} \right| \tag{17}$$

$$\left| Z_{cr.Wi} \right| = \left| \frac{(E_{th} / \sqrt{2[1 - \cos(\theta - \varphi)]})^2}{E_{th}^2 / 2Z_{th}[1 - \cos(\theta - \varphi)]} \right| = \left| Z_{th} \right| \tag{18}$$

From (17) and (18), it can be observed that, whether at a load node or a renewable energy node, the voltage satisfies the impedance matching condition at the critical instability point. Taking the load as an example, the normalized impedance margin index can be expressed as:

$$K_{r.Li} = \frac{\left| Z_{cr.Li} \right| - \left| Z_{th} \right|}{\left| Z_{cr.Li} \right|} = 1 - \frac{\left| Z_{th} \right|}{\left| Z_{cr.Li} \right|} \tag{19}$$

When the magnitude of the equivalent impedance Z_L of the load is greater than the magnitude of the Thevenin equivalent impedance Z_{th} of the node, where $0 < K_{r.i} < 1$, the voltage at the node is stable. A higher value of $K_{r.i}$ indicates a greater voltage stability margin. When they are equal, the system is in a critically stable state, $K_{r.i} = 0$. From another perspective, a smaller magnitude of Thevenin equivalent impedance Z_{th} implies a stronger capability of the node to handle the load (accept renewable energy injection). The magnitude of Thevenin equivalent impedance can to some extent reflect the static voltage stability of the node. The short-circuit ratio for a

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single-infeed system is an index based on this physical characteristic of Thevenin equivalent impedance. In fact, if we consider the base voltage of the AC system as the rated voltage at the grid connection point and the base power as the rated power of the load (renewable energy), then the short-circuit ratio at the grid connection point can be expressed as:

$$S_{CR}=\frac{S_{ac}}{P_N}=\frac{U_N^2/Z_{th}}{P_N}=\frac{1}{Z_{pu}} \tag{20}$$

In (20), S_{ac} represents the short-circuit capacity of the node, P_N and U_N is the rated power and rated voltage of the node, and Z_{pu} is the normalized magnitude of Thevenin impedance. It can be observed that the short-circuit ratio index can also be used for qualitative assessment of static stability, and it possesses a certain degree of reasonability.

Taking the single-machine black-box system shown in Fig 1 as an example, considering Thevenin equivalent voltage denoted as 1(p.u) and Thevenin equivalent impedance denoted as 0.01+j0.1(p.u), with power flow out of the system as the reference direction, let's analyze the characteristics of the margin indices for two scenarios: load and renewable energy. The initial power for the load is represented as 2+j0.5(p.u), and for the renewable energy, it is represented as -2-j0.5(p.u).

Let the power of both the load and the renewable energy source increase while maintaining a constant power factor until the system becomes unstable. The variation of the upper branch of the P-V curve and the margin indices during the instability process is illustrated in Fig 2.

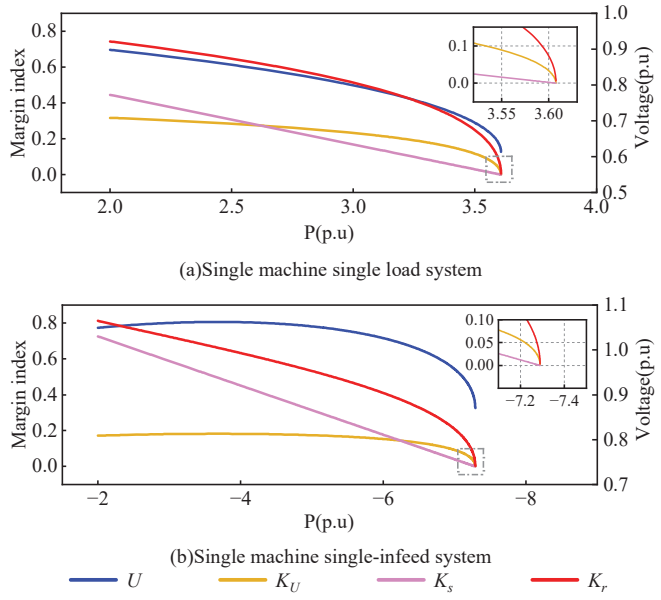


Fig. 2. The margin indices and voltages of the single-machine black-box system

From the results in Fig 2, it can be observed that as the transmission power continuously increases, the values of the three stability margin indices decrease. When the system reaches critical stability, all three indices decrease to zero, indicating the collapse point of voltage. The critical voltage and critical power of the load are both lower than those of the new energy generation station, consistent with the earlier analysis results. Further analysis of the characteristics of the three indices reveals that the power margin index K_P has the best linearity, with the highest accuracy near instability. The impedance margin index K_r, as it doesn't require calculating the Thevenin potential, has the easiest computation process but exhibits the poorest linearity near instability. The voltage margin index K_U is the most intuitive but shows a situation of initial increase followed by a decrease.

Maintaining a constant power factor growth, simulations were conducted with R_{th} values of 0 and 0.02, respectively. The critical stability conditions, critical short-circuit ratio, and various indices of the system at the critical stability point are calculated under these conditions, as shown in Table I.

TABLE I. THE VALUES OF EACH INDEX IN THE SINGLE-MACHINE BLACK-BOX SYSTEM AT THE CRITICAL STABILITY POINT

Index	R_{th}	K_U	K_S	K_r	S_{cr}
Load	0.00	0.0048	3.6×10^{-5}	0.038	2.43
	0.01	0.0037	2.7×10^{-5}	0.036	2.76
	0.02	0.0031	2.3×10^{-5}	0.032	2.95
Source	0.00	0.0039	4.1×10^{-5}	0.011	1.56
	0.01	0.0044	6.7×10^{-5}	0.013	1.36
	0.02	0.0065	9.8×10^{-5}	0.025	1.18

Comparing the stability margin indices and the short-circuit ratio indices at critical stability, it can be observed that the critical short-circuit ratio indices vary significantly under different operating conditions, making it difficult to establish a uniform standard for determining instability. In contrast, the margin indices are consistently zero at the critical stability point, regardless of the operating conditions, indicating a more universal significance.

THE APPLICATION OF MARGIN INDICES IN THE STATIC VOLTAGE STABILITY ANALYSIS OF RENEWABLE ENERGY MULTI-INFEED SYSTEMS

Centralized distributed renewable energy stations typically exhibit a multi-infeed configuration. Due to the proximity of these renewable energy stations, they can have mutual influences. Some studies have proposed direct research based on the topology of renewable energy multi-infeed systems, either decoupling them into single-infeed systems based on mathematical theories [11] or quantifying the influences between renewable energy stations directly [12].

In fact, the form of multi-infeed systems does not affect the application of Thevenin equivalent methods. For example, when studying the voltage stability of a particular renewable energy station, the remaining renewable energy stations can still be equivalently represented on the system side, as shown in Fig 3. The equivalent parameters obtained in this case already incorporate the influences of the other renewable energy stations on the equivalent nodes. Therefore, research can still be conducted based on the topological structure illustrated in Fig 1.

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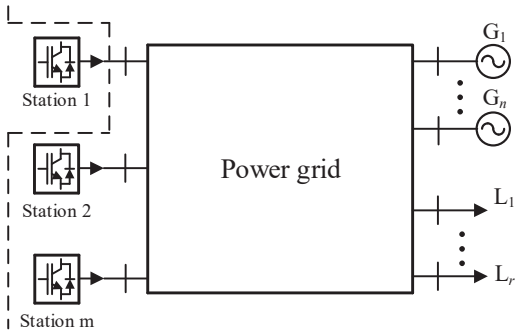


Fig. 3. Equivalent diagram of renewable energy multi-infeed system

From the above analysis, it can be observed that in the process of evaluating stability by equating a multi-feed system to a single-feed system, the accurate calculation of margin indices critically depends on the precise computation of the Thevenin parameters. Since the Thevenin equivalent involves simplifying the circuit topology, it inevitably omits some electrical information that is deemed less relevant while retaining essential details. Reference [9] employs a method based on the transformation of nodal admittance matrices to compute Thevenin parameters. However, this method overlooks the nonlinear processes of loads and synchronous generators, introducing a certain degree of accuracy concern. Furthermore, due to the strong nonlinearity of the impedance margin index near the instability point, the accuracy of Thevenin parameters becomes even more crucial.

In static voltage stability analysis, the voltage-to-power derivative (also known as static sensitivity) is a crucial variable, representing the slope at each point on the P-V curve. During the transition from stable to critically stable conditions in a system, the voltage-to-power derivative undergoes a rapid change from a small value to infinity, exhibiting a distinctive and highly indicative characteristic. Therefore, to better preserve the information of the original system's P-V curve during the instability process, maintaining the invariance of the voltage-to-power derivative can be used as a calculation condition for equivalent parameters. This Thevenin equivalent method is more suitable for static voltage stability analysis, and the specific parameter calculation process can be found in reference [15].

Since the voltage-to-power sensitivity parameters are dynamically obtained through the Jacobian matrix during the power flow calculation, they inherently encapsulate changes in various power sources and loads within the system. Therefore, the accuracy of the obtained Thevenin equivalent parameters is high. Based on these Thevenin equivalent parameters and the current information on power and voltage for renewable energy sources, various margin indices can be calculated. This allows for the assessment of static voltage stability in the context of a renewable energy multi-infeed system.

IV. Case Study

This section will validate the effectiveness of three margin indices in a renewable energy multi-infeed system through simulation. A modification is applied to the IEEE 39-node system by converting synchronous generators 32, 33, and 34 into wind power stations with constant power factor control. In the power flow analysis, these three nodes are treated as PQ nodes. The modified system is illustrated in Fig 4.

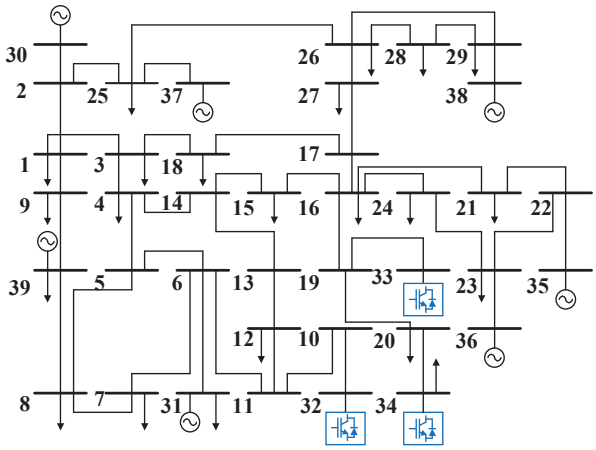


Fig. 4. Modified IEEE 39-node system

During the investigation of the increasing power of the wind power stations, two power balancing methods are employed. Method 1 involves an increase in the overall load power while keeping the active power of synchronous generators constant. Method 2 entails a decrease in the output of all synchronous generators while maintaining the load and renewable energy power constant. This is done to validate the effectiveness of the three indices under different growth scenarios. As the research on margin indices on the load side is more mature, this section focuses solely on the calculation of margin indices on the renewable energy side.

A. Power balancing method 1

Let the wind power station at node 32 grow according to the manner described in (21) while keeping the active power output of the other generators unchanged. Throughout the growth process, maintain the power factor of the station constant. The incremental power of the wind power station is balanced by a proportional increase in the overall load. This continues until the system becomes unstable.

$$\begin{cases} P_{W.32}(\lambda) = (1+\lambda) P_{W.320} \\ Q_{W.32}(\lambda) = (1+\lambda) Q_{W.320} \end{cases} \tag{21}$$

In (21): λ represents the power growth coefficient; $P_{W.320}$ is the initial active power of wind power station 32; $Q_{W.320}$ is the initial reactive power of wind power station 32.

Perform Thevenin equivalence for node 32, and calculate the changes in three types of indices for node 32 as the system gradually becomes unstable. For wind power stations at nodes 33 and 34, conduct similar power increases. The changes in the three types of margin indices and node voltage during the system's instability process are shown in Fig 5

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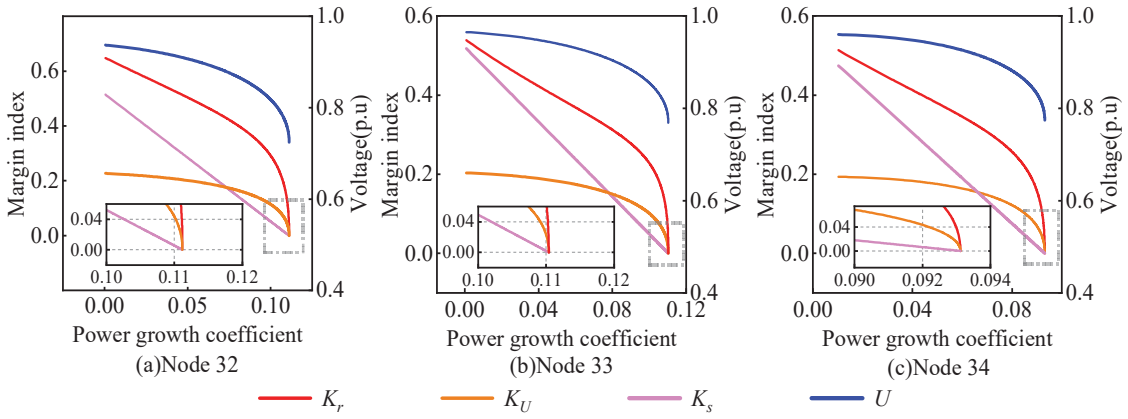


Fig. 5. Indices and voltage curves in power balance mode 1

In addition to verifying the effectiveness of the validation indices in the renewable energy multi-infeed system, this article also compares the accuracy of the indices obtained using the adopted equivalent method and the method in reference [9]. The change curves of the impedance margin index for both methods during three consecutive power flow processes are calculated separately, as shown in Fig 6.

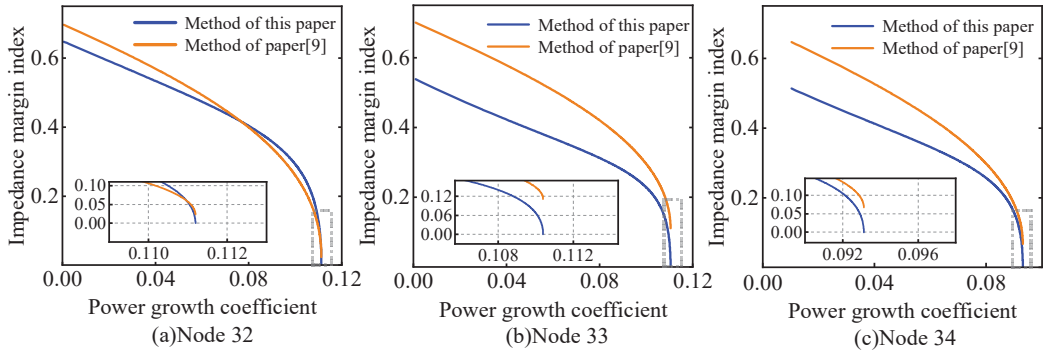


Fig. 6. Impedance margin index of each node in power balance mode 1

The node voltages and various indices under partial power flow conditions during the system instability process are shown in Table II

TABLE II. PART OF THE INDEX AND NODE VOLTAGE IN POWER BALANCE MODE 1

Node	λ	Voltage	K_U	K_S	K_r(this paper)	K_r(paper[9])
32	0.0000	0.9370	0.2269	0.5141	0.6477	0.6964
	0.0746	0.8772	0.1742	0.1703	0.4243	0.4271
	0.1002	0.8174	0.1136	0.0059	0.2903	0.2578
	0.1112	0.7378	7.4×10^{-3}	1.42×10^{-4}	0.0011	0.0233
33	0.0000	0.9650	0.2035	0.5178	0.5387	0.7022
	0.0793	0.9052	0.1509	0.1470	0.3146	0.4236
	0.1025	0.8452	0.0906	0.0373	0.2100	0.2725
	0.1104	0.7794	6.2×10^{-3}	1.10×10^{-4}	0.0182	0.1115
34	0.0000	0.9598	0.1934	0.4745	0.5134	0.6466
	0.0736	0.9002	0.1398	0.1122	0.2910	0.3473
	0.0888	0.8394	0.0783	0.0245	0.1763	0.2005
	0.0931	0.7842	7.2×10^{-3}	1.49×10^{-4}	0.0188	0.0782

B. *Power balancing method 2*

Assuming that 30% of the active power increment is balanced by the balanced nodes, and the remaining 70% is balanced by the synchronous generators of PV nodes in proportion to their active power output. The reactive power of the wind power station also increases accordingly to maintain a constant power factor.

Similarly, the variations of three margin indices and node voltages during the system instability process are shown in Fig 7.

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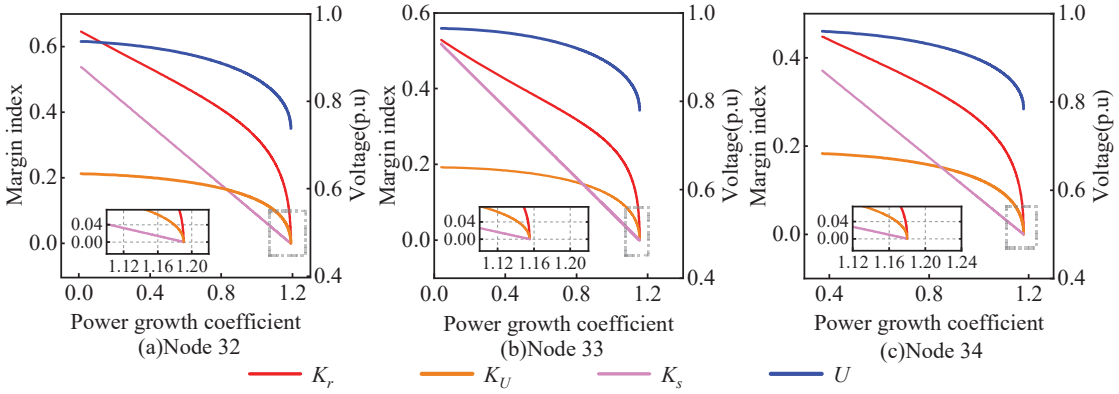


Fig. 7. Indices and voltage curves in power balance mode 2

Similarly, a comparison between the method employed in this paper and the method proposed in reference [9] is illustrated in Fig 8. Node voltages and various indices under partial power flow conditions during the system instability process are presented in Table III.

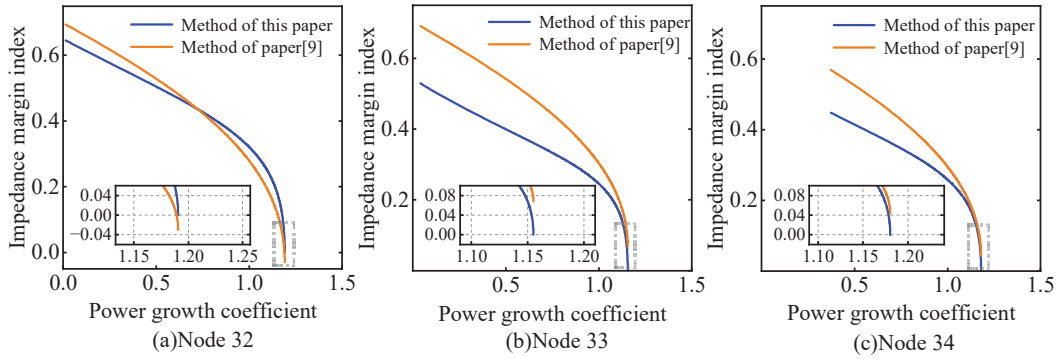


Fig. 8. Impedance margin index of each node in power balance mode 2

TABLE III. PART OF THE INDEX AND NODE VOLTAGE IN POWER BALANCE MODE 2

Node	λ	Voltage	K_U	K_S	K_r(this paper)	K_r(paper[9])
32	0.0000	0.9370	0.2126	0.5379	0.6458	0.6943
	0.8665	0.8772	0.1589	0.1480	0.3840	0.3640
	1.1066	0.8172	0.0972	0.0383	0.2400	0.1815
	1.1906	0.7430	7.0×10^{-3}	1.36×10^{-4}	0.0193	-0.0300
33	0.0000	0.9650	0.1923	0.5176	0.5290	0.6910
	0.8933	0.9052	0.1390	0.1215	0.2889	0.3708
	1.0978	0.8452	0.0779	0.0266	0.1775	0.2132
	1.1552	0.7794	4.9×10^{-3}	7.80×10^{-5}	0.0137	0.0678
34	0.0000	0.9598	0.1830	0.3708	0.4478	0.5692
	0.9836	0.9006	0.1287	0.0902	0.2661	0.3059
	1.1438	0.8402	0.0664	0.0167	0.1464	0.1587
	1.1801	0.7842	2.3×10^{-3}	1.5×10^{-5}	7.3×10^{-3}	0.0431

From Fig 5 and Fig 7, it can be observed that for the renewable energy multi-infeed system, under different power growth scenarios, all three margin indices are accurately indicative of system instability with a value of 0 at the critical stability point. This aligns with conclusions drawn from single-infeed systems. The results in Fig 6 and Fig 8 indicate that there are some differences in the margin indices obtained by different equivalent methods, and the method used in this paper has higher accuracy under both growth scenarios. Furthermore, a comprehensive analysis of the charts reveals that under different growth scenarios, the critical power growth coefficients for the same renewable energy unit may vary significantly (approximately 10 times in the two growth scenarios used in this paper). In practical applications, it is essential to analyze both margin indices and the system power growth process comprehensively.

CONCLUSION

This study proposes the use of the voltage stability margin index to analyze the static voltage stability of the system from the perspective of renewable energy power stations. A normalized expression of the index is derived based on a single-machine black-box system, extending its application scenarios. The main contributions are as follows:

The normalized expressions of the index on the load and renewable energy sides are derived. The characteristics of three

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different indices are analyzed and compared with the commonly used short-circuit ratio index, effectively avoiding the difficulties in calculating the critical short-circuit ratio index.

The effectiveness of the margin index's evaluation is influenced by the accuracy of the calculation of Thevenin equivalent parameters. In the calculation of Thevenin equivalents, the proposed approach assumes that the sensitivity of voltage to power remains unchanged before and after equivalence. This method of equivalence is more suitable for static stability evaluation compared to commonly used methods in existing literature, yielding more accurate results.

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